

Program 7

Design and implement C/C++ Program to solve discrete Knapsack and continuous Knapsack problems using greedy approximation method

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    float weight[50],profit[50],ratio[50],Totalvalue,temp,capacity,amount;
```

```
    int n,i,j;
```

```
    printf("Enter the number of items :");
```

```
    scanf("%d",&n);
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        printf("Enter Weight and Profit for item[%d] :\n",i);
```

```
        scanf("%f %f", &weight[i], &profit[i]);
```

```
    }
```

```
    printf("Enter the capacity of knapsack :\n");
```

```
    scanf("%f",&capacity);
```

```
    for(i=0;i<n;i++)
```

```
        ratio[i]=profit[i]/weight[i];
```

```
    for (i = 0; i < n; i++)
```

```
        for (j = i + 1; j < n; j++)
```

```
            if (ratio[i] < ratio[j])
```

```
            {
```

```
                temp = ratio[j];
```

```
                ratio[j] = ratio[i];
```

```
                ratio[i] = temp;
```

```
                temp = weight[j];
```

```
                weight[j] = weight[i];
```

```
                weight[i] = temp;
```

```
                temp = profit[j];
```

```
                profit[j] = profit[i];
```

```
                profit[i] = temp;
```

```
            }
```

```
    printf("Knapsack problems using Greedy Algorithm:\n");
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        if (weight[i] > capacity)
```

```
            break;
```

```
        else
```

```
        {
```

```
            Totalvalue = Totalvalue + profit[i];
```

```
            capacity = capacity - weight[i];
```

```
        }
```

```
    }
```

```
    if (i < n)
        Totalvalue = Totalvalue + (ratio[i]*capacity);

    printf("\nThe maximum value is :%f\n",Totalvalue);
    return 0;
}
```

Sample Input and Output:

Enter the number of items :5

Enter Weight and Profit for item[0] :
10 25

Enter Weight and Profit for item[1] :
20 60

Enter Weight and Profit for item[2] :
30 40

Enter Weight and Profit for item[3] :
40 35

Enter Weight and Profit for item[4] :
50 5

Enter the capacity of knapsack :
60

Knapsack problems using Greedy Algorithm:

The maximum value is :125.000000